

EKL Safety Training

Safety is the responsibility of everyone, everywhere, anytime.



December 2025 - Standard EKL version

Introduction

During this course you will be informed about the safety rules and precautions in the EKL cleanrooms. You must learn how to work safely, how to act when an incident or accident happens, and what to do in case of an alarm.

- The safety of the lab and its users are the most important.
- You must always think about safety with every action that you take.



Safety risks

Working in a cleanroom or laboratory results in a higher safety risk than working in an office environment:

Many chemicals are being used:

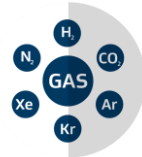
- Solids



- Liquids



- Gases



Some of these are **hazards** !!!

EKL safety measures

EKL must comply to the legislation from the Government and the City of Delft, and has the following safety measures:

- A Safety Officer and a Safety Committee.
- An Emergency Response Team (in Dutch: BHV = Bedrijfs Hulp Verlening).
- Education and training of users.
- An Alarm System with Alarm Buttons and a Gas Detection system.
- An Access Control system.

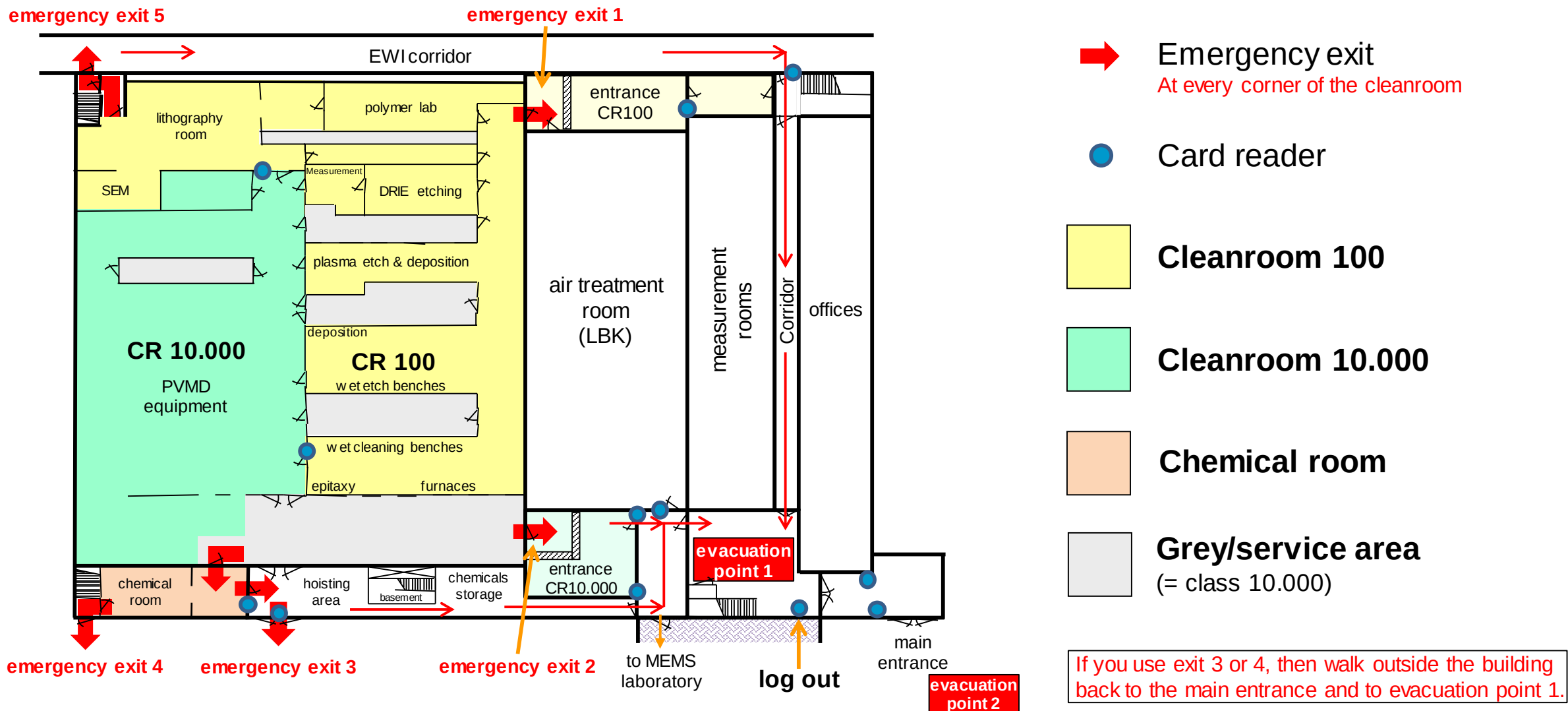


Requirements to work in the labs

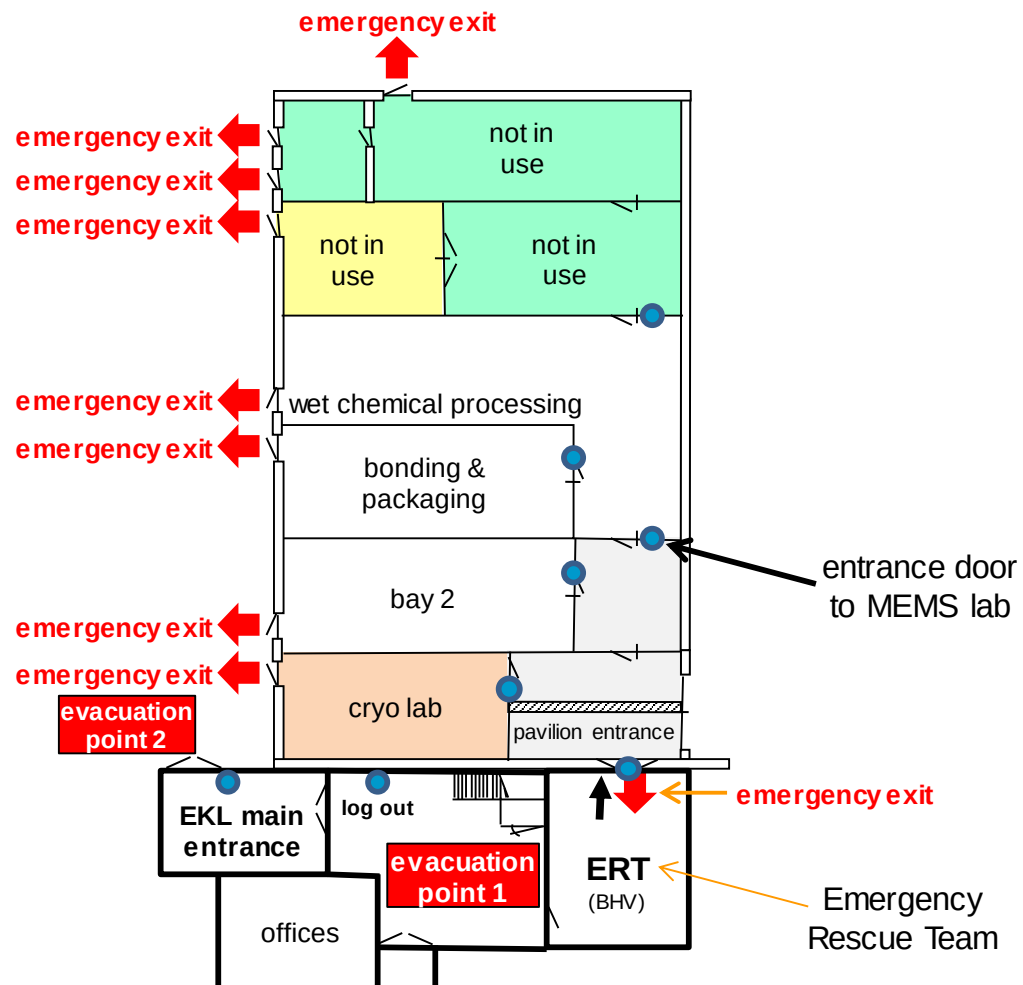
- Become familiar with the **map of the labs**.
- **Know where to find:**
 - emergency exits
 - alarm buttons
 - fire extinguishers
 - emergency kits
 - showers
- **Keep access free to:**
 - escape routes
 - fire extinguishers
 - emergency kits
- Always follow the instructions of the **Emergency Response Team (ERT)**.

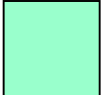
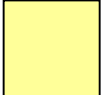


CR100/CR10000 escape routes in case of an alarm



Pavilion/MEMS escape routes in case of an alarm



-  Emergency exit
-  Card reader
-  Currently not in use
-  Litho room
-  Cryo laboratory

When ending up outside the building, then walk back to the main entrance and go to evacuation point 1.

Emergency Response Team

The ERT consists of trained staff members who will help with incidents ranging from personal injuries and chemical incidents, to extinguishing a fire and rescuing people.

ERT members have this sign on their office



Access control

The access control system:

The main purpose of access control is to **provide security** to EKL users and visitors by allowing or restricting access to the labs.

The following rules apply:

- **Sign in:** Use your card to enter the EKL building, the labs and other restricted areas.
- **Log out:** Use your card when leaving the labs, restricted areas and the EKL building.

In case of an alarm or incident the Emergency Response Team and/or the Fire Brigade can check who is still where in the EKL building.



Type of alarms

- **Automatic smoke detector alarm**

This activates an alarm sound, and on the alarm panels only the text "**Cleanroom verlaten**" (= leave cleanroom) will light up.



"Cleanroom/Paviljoen verlaten" sound

- **Automatic gas detector alarm**

This activates an alarm sound, and on the alarm panels both the text "**Gasalarm**" and "**Cleanroom verlaten**" will light up.



"Gasalarm" sound

- **Manual alarm (pressed button)**

This activates an alarm sound, and on the alarm panels only the text "**Cleanroom verlaten**" will light up.



During an alarm a warning signal sounds and the panel will be (partly) illuminated.

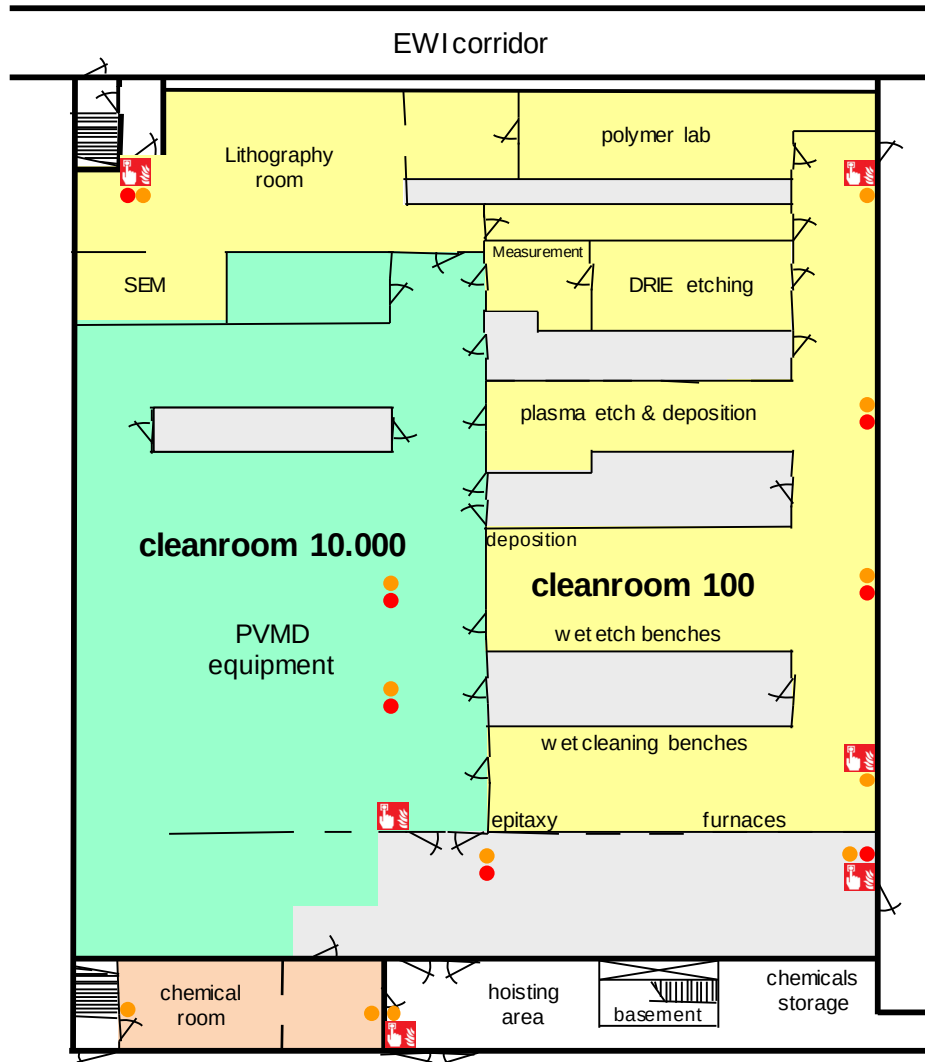
An alarm test is performed every first Monday of the month at 8:30 hrs.

Function of the alarm buttons in EKL

Name	Picture	When	Result	Additional
Calamity button		If you need immediate help from the ERT (e.g. accident with a person, chemical spill).	- An alarm sounds and the sign "Cleanroom verlaten" lights up. - The ERT is alarmed. - The lab must be evacuated. - All standard access doors will be unlocked.	If somebody is injured then provide help until the ERT arrives. Always think of your own safety!
Emergency button		In case of a gas leak or other severe threat to the lab and its users (e.g. large chemical spill generating dangerous fumes).	- An alarm sounds and the sign "Cleanroom verlaten" lights up. - The ERT is alarmed. - All labs and basement must be evacuated. - All standard access doors will be unlocked. - All gas valves are closed (except N2). - Air recirculation stops: Only fresh air goes in.	If somebody is injured then provide help until the ERT arrives. Always think of your own safety!
Fire Alarm Button		In case of fire.	- An alarm sounds and the sign "Cleanroom verlaten" lights up. - The ERT is alarmed. - All labs and basement must be evacuated. - All standard access doors will be unlocked. - All gas valves are closed (except N2). - Air recirculation stops: Only fresh air goes in. The Fire Department is alarmed.	If safe and possible (for waste basket size fires), try to extinguish the fire with a fire extinguisher. If extinguishing is not possible or no longer safe, then stop and leave the room. Please note that a CO₂ extinguisher only lasts about 30 seconds. Leave the room immediately if you don't know how to use a fire extinguisher!
EMO button		In case of a dangerous situation with a machine (e.g. smoke comes out, a person is trapped inside system, liquid is leaking).	No cleanroom alarm is generated. - Only for this tool: System will switch off (i.e. no electricity is present in the system after the main power switch).	Warn the tool owner.

Important: Inform the ERT at the evacuation point why you pressed the alarm.

Location of the alarm buttons in CR100/CR10000



● Calamity button



● Emergency Stop button

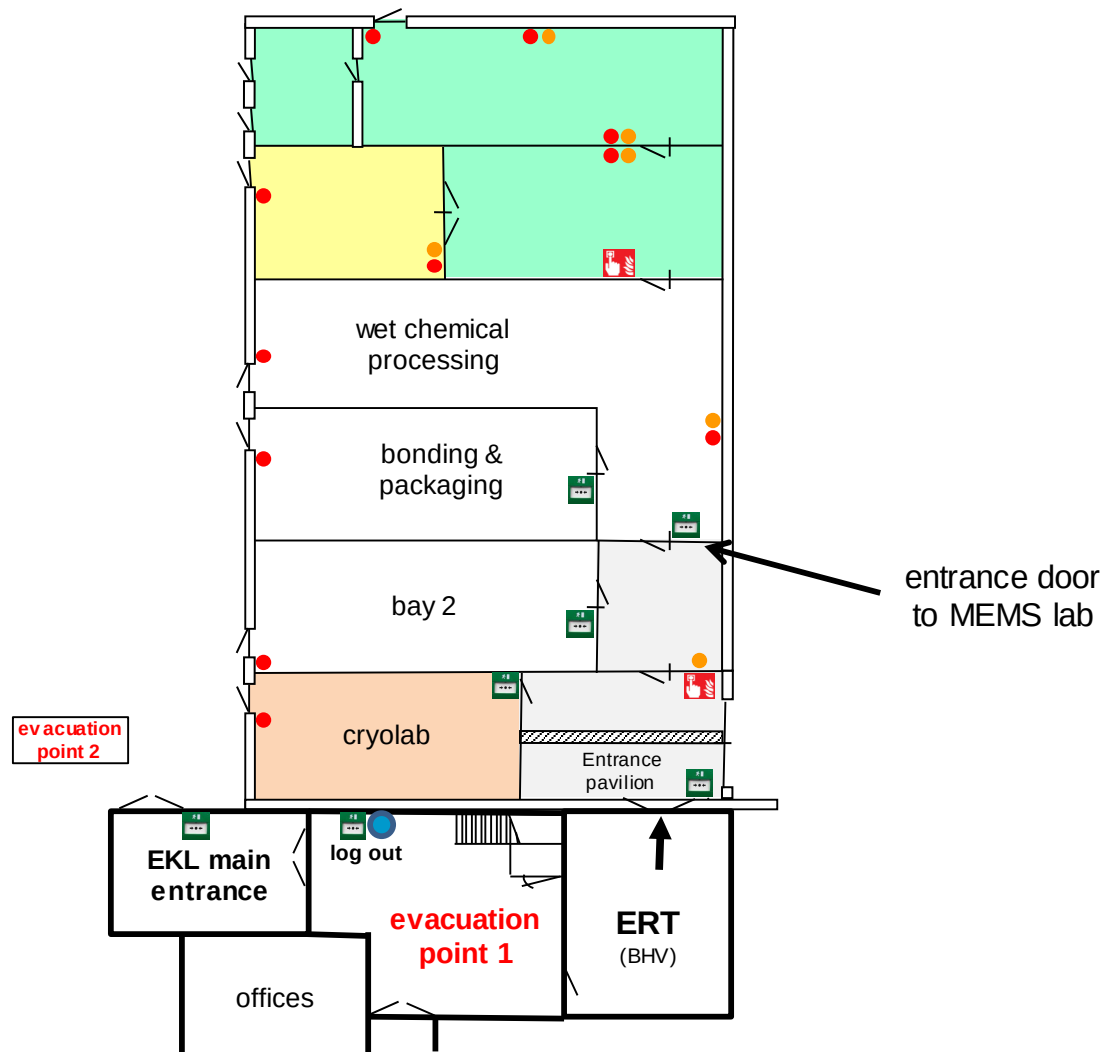


Fire Alarm button
(usually next to an exit door)



Be aware where the alarm buttons are located!

Location of the alarm buttons



● Calamity button



● Emergency Stop button



■ Fire Alarm button
(usually next to an exit door)

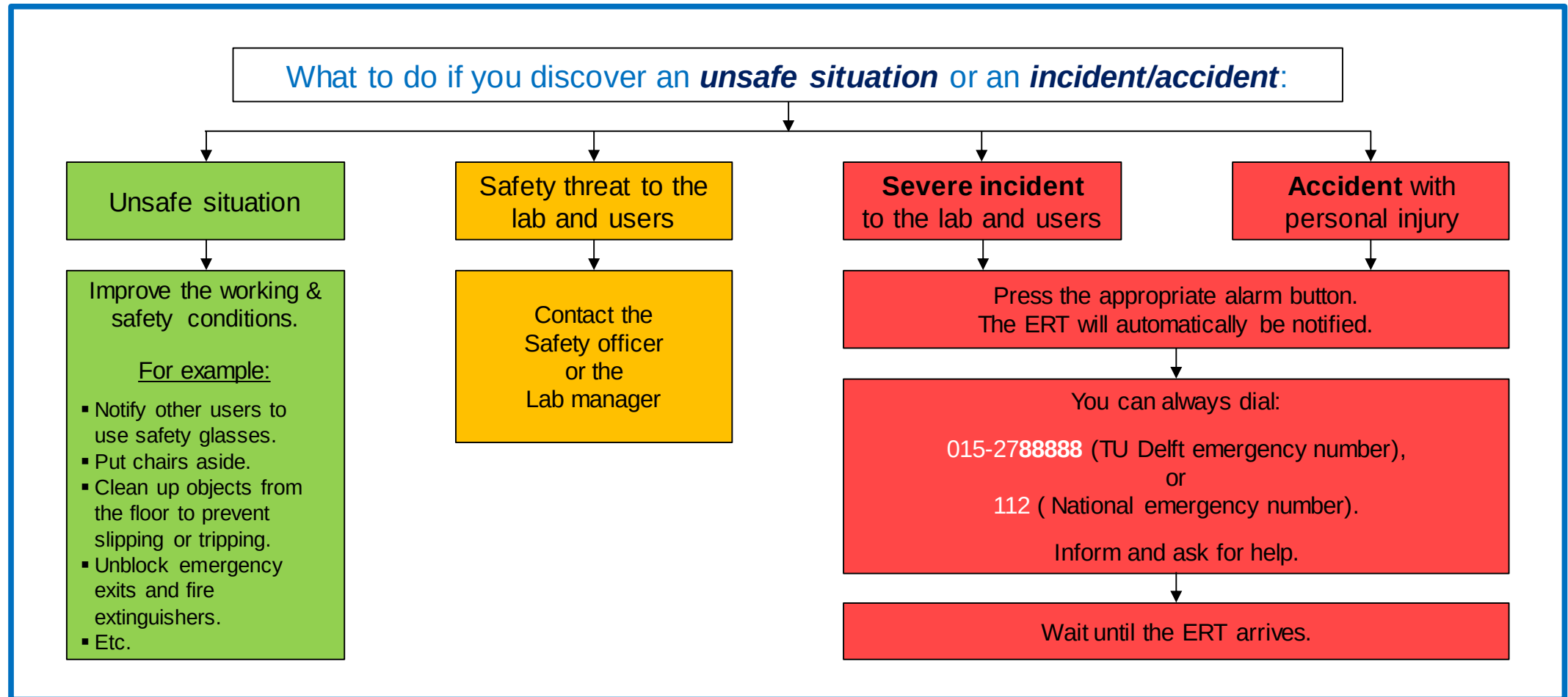


■ Door unlock button
(if not automatically unlocked)



Be aware where the alarm buttons are located!

Overview of safety procedures



All safety incidents and accidents must be reported on SharePoint.

What to do if there is an alarm

- Leave the work area immediately. Do not finish your process.
- Alert other users on your way out.
- Follow the escape route to the nearest emergency exit.
 - All doors will automatically unlock.
(If not, then press the button in the green box next to the door.)
 - Close the doors behind you.
 - Do **NOT** take off your cleanroom suit or lab coat.
- Go to evacuation point 1 and log out.
 - Here users can take off their cleanroom suit or lab coat.
- Wait for further instructions.

Note:

- Evacuation point 2 is only used if point 1 is unsafe.
- There is also a 3rd evacuation point that is located on the Mekelpark. This point will only be used if both other points are unsafe.

If you know what caused the alarm then inform the ERT.



Sign indicating the escape route



Green box to unlock the door
(if not automatically unlocked)



Evacuation point 1

inside the building at
the reception hall



Evacuation point 2

outside the building in front of
the main entrance

Use of chemicals during lab opening hours

- **Standard opening hours: CR + MEMS ⇒ Monday - Friday 8:00 - 18:00, except holidays**

It is **ALLOWED** to transport, pour, mix and dispose (approved) chemicals between 08:00 - 17:00.

- For working with the CR100 wet benches: A buddy is required.
(these bathes are already prepared)
- For **handling** chemicals: Your buddy must be **in the same area**.
(handling includes transport, pouring, mixing, disposing)

- **Extended opening hours: CR ⇒ Monday - Friday 18:00 - 22:00, weekends and holidays**
MEMS ⇒ Monday - Friday 18:00 - 20:00, weekends and holidays

It is **NOT ALLOWED** to transport, pour, mix and dispose chemicals, also **NOT** when a buddy is present.

- For working with the CR100 wet benches: A buddy is required.
(these bathes are already prepared)

⊗ Violation of the above rules may result in **revoked access rights**.

Personal protection when working with chemicals

- Everyone needs a **buddy** to work with chemicals or with liquid Nitrogen.
- Wear **safety glasses**. (If you already wear normal glasses, then over your normal glasses), or a **protective face shield** (also when you are looking over someone's shoulder).
- Wear **neoprene gloves** over the white nitrile cleanroom gloves.
 - Check if the Neoprene gloves are in a good condition.
 - If the condition is not OK then throw them away and get new ones.
- Use an apron to protect your body when you are pouring chemicals.
- Work in a fume hood or wet bench.



Additional info for working with chemicals

Mandatory

- Wear trousers and closed shoes under your cleanroom suit. (Shorts and open shoes are not allowed).
Cleanroom suits do not protect you against chemical spills. They only protect the cleanroom environment against the dust that you produce. Your clothes are the only absorbent in case of a spill.
- Rinse the neoprene gloves regularly with water during the mixing/pouring and when your ready.

Advices

- Wearing contact lenses while working with chemicals is not advised.
If you get liquids or fumes into your eyes they may diffuse behind the lenses or react with them.
- It is strongly advised to wear clothes that cover your arms when working with chemicals.
Note: Protective disposable arm sleeves are available at EKL.



What to do in case of a chemical spill

We distinguish between 3 types of chemical spills:

1. A minor chemical spill outside a fume hood or a wet-bench:
a few drops
2. A major chemical spill outside a fume hood or a wet-bench:
e.g. a falling beaker/bottle containing chemicals, or a leak of chemicals
3. A chemical spill on a **person**:
any size of spill on your hand, face or body

What to do in case of a minor chemical spill outside a fume hood or a wet-bench ⇒ a few drops

How to react:

- Think about your own safety and avoid breathing in vapors from the spill.
- Alert people in the immediate area of the spill and secure the area (so nobody gets contaminated or slips over the spilled chemical).
- Put on neoprene gloves.
- Use a wet blue towel to clean up the drops.
- Use a new blue towel to dry the wet area.
- Inform the Wet Bench team about what happened.

What to do in case of a major chemical spill outside a fume hood or a wet-bench ⇒ a falling beaker or bottle, or a leak of chemicals

How to react:

- Press the **CALAMITY** button.
- Think about your own safety and avoid breathing in vapors from the spill.
- Evacuate the laboratory and warn people to avoid that someone gets contaminated or slips over the spilled chemical.
- Follow the [What to do if there is an alarm](#) procedure. (Click on the link to see that slide.)
- Inform the ERT at the evacuation point.
- Report the incident on SharePoint.

What to do in case of a chemical spill on a person

Reaction as a victim:

- Press the **CALAMITY** button.
- Try not to panic.
- Ask another user for help (buddy).
- Move away from the splashed chemical, if possible into a safe area.

Reaction as a helper:

- Press the **CALAMITY** button.
- Always think about your own safety.
- Stay with the victim and try to help.
- Ask another user for help (buddy).
- Use neoprene gloves from the emergency kit.
- Move the victim away from the splashed chemical into a safe area if possible.

In both cases:

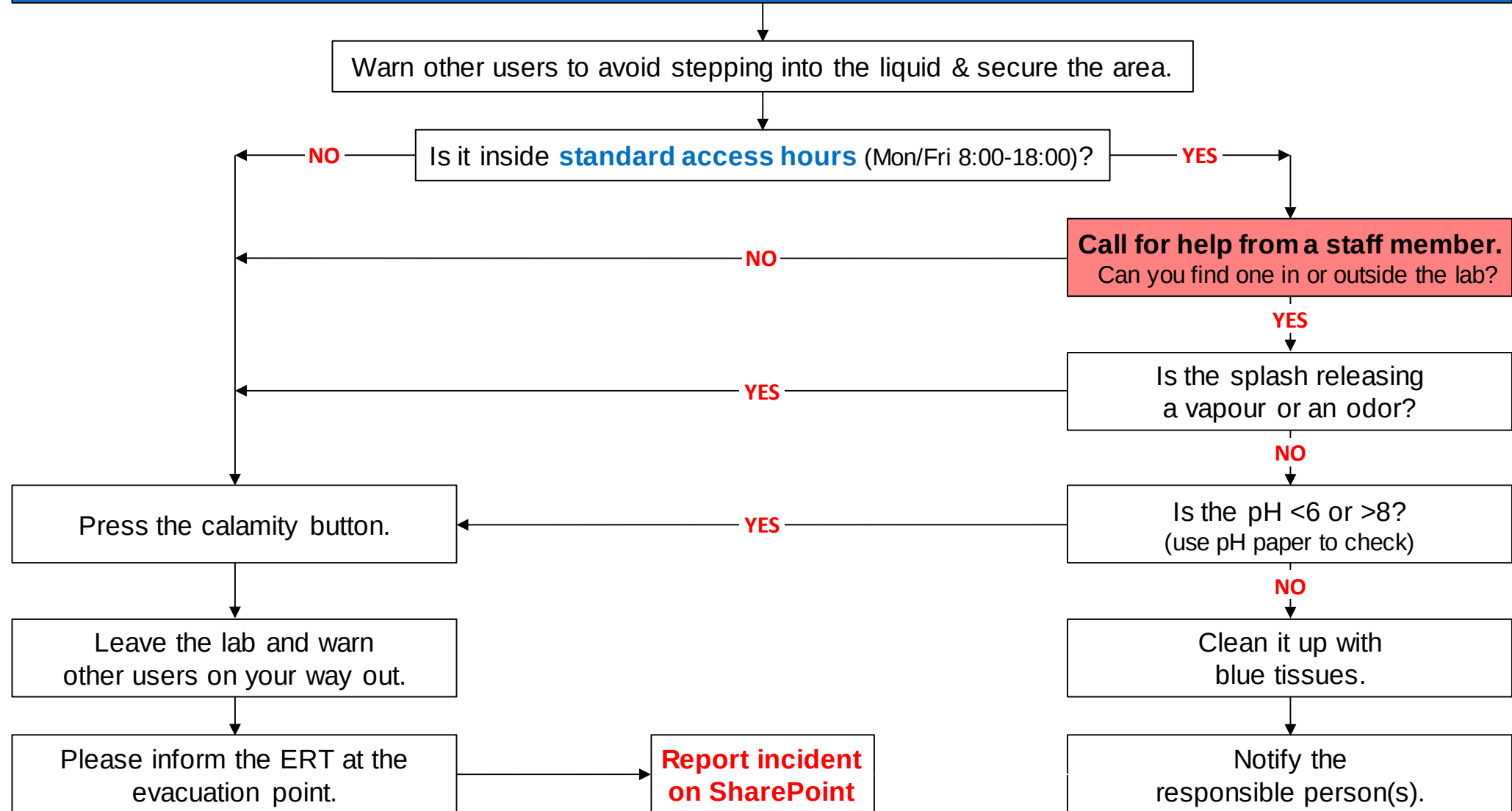
- Apply the appropriate neutralizing solution on the victim (on clothes if necessary):
 - Organic contamination (e.g. for acetone, NMP, IPA): → Use Diphoterine or Water.
 - Acid or a Base contamination (e.g. nitric acid, KOH, TMAH): → Use Diphoterine.
 - HF contamination (e.g. Poly-Si etch, BHF, HF): → Use Hexafluorine.
 - If you get a drop in your eye: → Use the eye bottle to flush. Use the whole bottle, but do not squeeze.
- Remove the victims clothing, including boots & shoes (use scissors if necessary ➡ available in rescue kit). Try to prevent further spreading out of the chemical over the victim.
- Wait for help from the ERT.
- Report the accident on SharePoint.

How to handle in case of an incident/accident during **extended access hours**

1. Press the appropriate emergency button.
2. In case of a *severe incident to the lab and users* or in case of an *accident with personal injury*:
 - Dial: **015-2788888** TU Delft Emergency number.
 - Dial: **112** National Emergency number.
3. Supply the following information to the operator:
 - Your name.
 - The address of EKL ([Feldmannweg 17, 2628CT Delft](#)).
 - Location of the incident (Office, CR100, CR10000, MEMS, basement, etc.).
 - The nature of the incident (e.g. hazardous materials which are involved).
 - If an ambulance is needed (to your judgement).
4. Do **NOT** go home:
 - Wait for the ERT to arrive, this can take up to 20 minutes.
 - Supply them with detailed information about the incident or accident.

Extended opening hours: CR ⇒ Monday - Friday 18:00 - 22:00, weekends and holidays
MEMS ⇒ Monday - Friday 18:00 - 20:00, weekends and holidays

What to do if you find an unidentified liquid splash on the floor?



When is an acid or base weak or strong?

WEAK?

STRONG?

Weak or strong acids/bases and pH

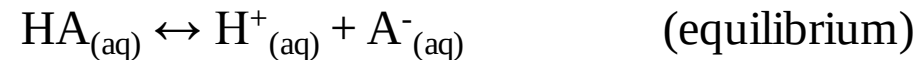
When an acid or base is dissolved in water, the splitting up of the molecules into ions is called:
dissociation reaction

The amount of molecules that split into ions defines if an acid or base is weak or strong:

- An acid or base is strong if **all** the molecules in the solution are dissociated.



- An acid or base is weak if only **a part** of the molecules is dissociated.
The rest of the molecules stay as they are and will not split up in the in solution.



Acid, Base or organic chemical spill

Use the Diphoterine-kit after a spill with acid, base or organic chemicals on a person.

- Think about your own safety.
- The best results are obtained when used <1min after the spill.
- Create an accident report on SharePoint.

What is it doing?

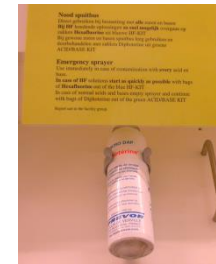
- Effective rinsing of the chemical product on the surface of the skin or the eye.
- The Diphoterine® solution will limit the penetration of the chemical product into your skin or eye.
- The Diphoterine® solution neutralizes the acid ions (H^+) and hydroxide ions (OH^-) very quickly and simultaneously.

Contents:

- Clean neoprene gloves.
- Diphoterine bags (with eye cap to use it as eye shower).
- Scissors.

Usage:

- Use a spray bottle to neutralize the contaminated body parts or clothes.
- Open the Diphoterine kit and put the neoprene gloves on (over your nitrile gloves).
- Remove the contaminated victims clothing (use the scissors if necessary).
- Open the Diphoterine bag: Rinse until the bag is empty without squeezing.



Spray bottle

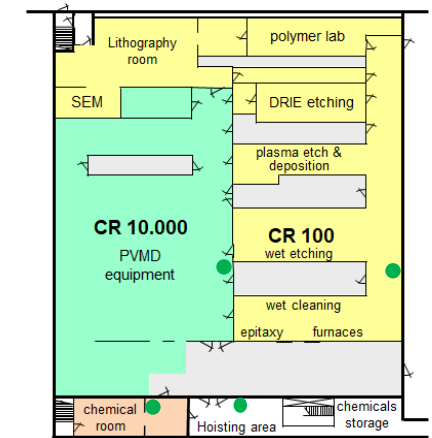


Diphoterine kit

Diphoterine spray bottles are located at all wet benches and fume hoods.



Eye cap to use the Diphoterine bag as eye an shower.

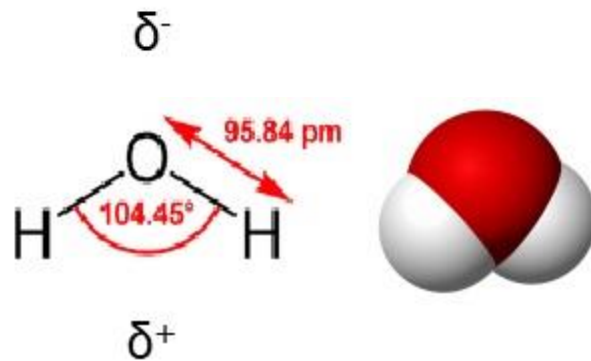


● = Location of Diphoterine kits

Question: Is HF, weakly acidic or dangerous?

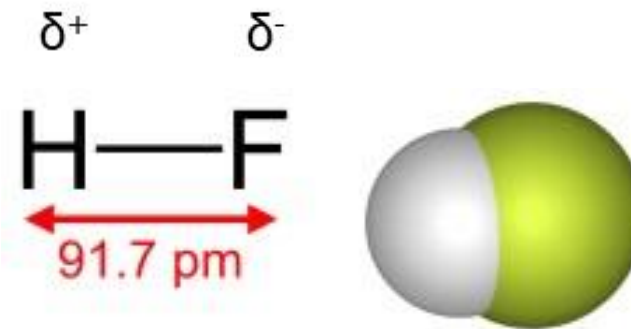
Molecule size

Water molecule



Size: 0.275 nm

HF molecule



Size: 0.091 nm

- Fluorine ions react with calcium and magnesium. This can cause:
- The death of cells which progressively leads to tissue necrosis
 - Heart arrhythmias and cardiac arrest

Hydrofluoric Acid (HF) spill

Use the HF-kit after a spill with Hydrofluoric acid on a person.

- Think about your own safety.
- The best results are obtained when used <1min after the spill.
- If needed go to hospital and bring a different, unopened HF-kit with you.
- Create an accident report on SharePoint.

What is it doing?

- Effective rinsing of the chemical product on the surface of the skin or the eye.
- The Hexafluorine® solution will limit the penetration of the chemical product into your skin or eye.
- The Hexafluorine® solution neutralizes the acid ions (H^+) and fluoride ions (F^-) very quickly and simultaneously.

Content:

- Clean neoprene gloves.
- Calcium Gluconate gel (for after-treatment).
- Medication for after-treatment in a hospital.
- Scissors.

Usage:

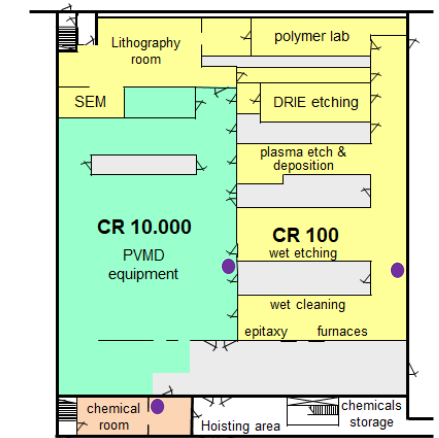
- Use a Diphoterine spray bottle to neutralize the contaminated body parts or clothes.
- Open the HF-kit and put the neoprene gloves on (over your nitrile gloves).
- Remove the contaminated victims clothing (use the scissors if necessary).
- Grab a Hexafluorine bottle: Rinse until the bottle is empty (use a 2nd one if needed).
- Apply Calcium Gluconate gel on the contaminated area.



Hexafluorine kit



Hexafluorine bottles



Organic chemical spill

When Diphoterine® is not available to treat an organic chemical spill, then use:

Eye shower bottles

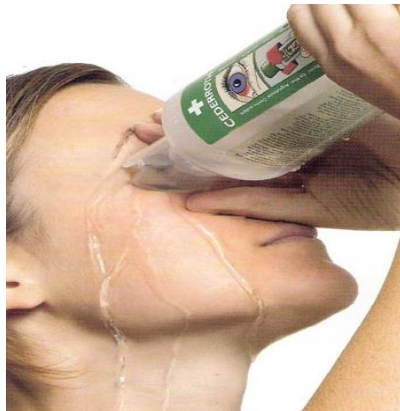
Use for:

- Washing organic chemicals or particles from your eyes.

Usage:

- Remove the seal from the bottle.
- Remove contact lenses (if possible).
- Keep eye open and wash eye (squeeze bottle, see picture).
- Flush for at least 15min.
- Use the entire bottle (or more).
- Create an incident report on SharePoint.

Bottle content: Sterilized water.



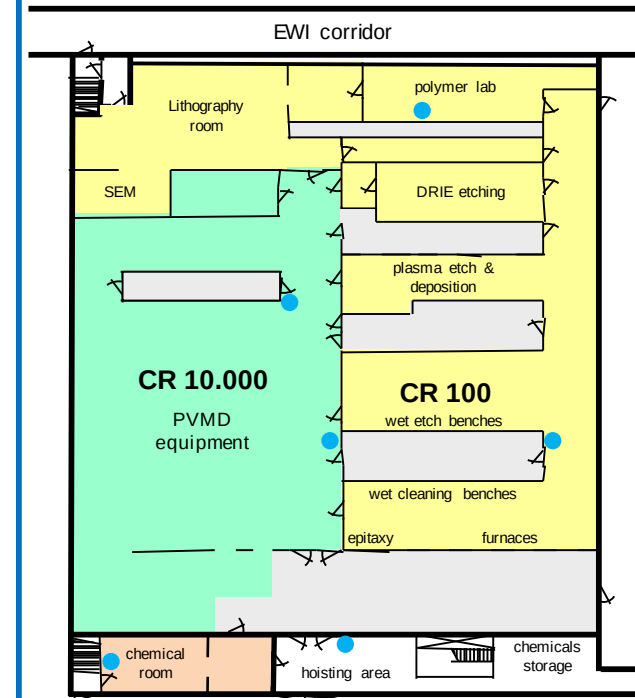
Water shower

Use for:

- Large organic chemical spills.
- Person on fire.

Usage:

- Stand under the shower.
- Pull the handle.
- Remove the victims clothing in case of a chemical spill, but **NOT** when a victim is on fire!
- Create an incident report on SharePoint.



● = Location of eye shower bottles or water showers

Danger of liquid Nitrogen (N_2)

Spillage or leakage of liquid N_2 can result in a lack of sufficient oxygen and cause respiratory problems.

1 liter of liquid N_2 produces **700 liters** of N_2 gas.

A room where liquid N_2 is being used has an O_2 detector that triggers an alarm if the Oxygen level is lower than 19%.

- If you see an unconscious person in a room with low oxygen content: **Do not enter.**
- Press the calamity button.
- Leave the area and alert people on your way out.
- At the evacuation point inform the ERT why you pressed the calamity button.
- Report the incident or accident on SharePoint (according to the protocol).

It is not allowed to use liquid N_2 without permission and Module training.



Liquid Nitrogen vessel



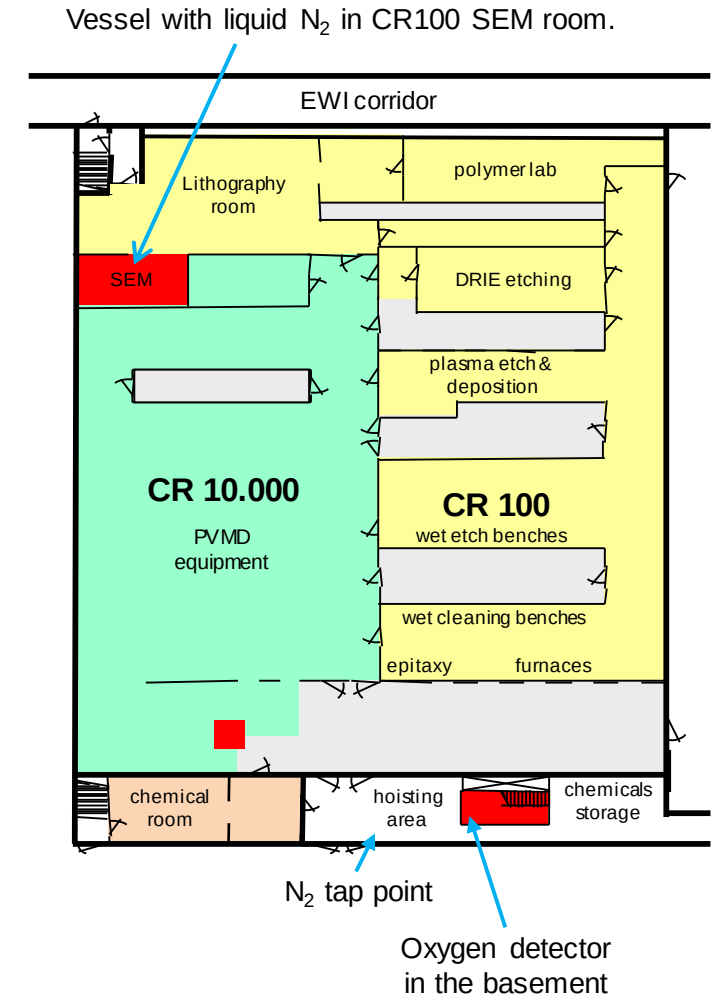
O_2 detector



O_2 detector in basement



N_2 tap point



Chemicals at EKL

Chemicals used at EKL:

- There are $\pm$ 300 different liquid and solid chemicals present.
- Many different gasses are used.
- Most of these chemicals are very hazardous.
- Most users will work with them and all users will be close to them.
- All of these chemicals are registered and have a MSDS.

It is important to know:

- How dangerous these chemicals are.
- How to use the personal protective equipment (fume hood or wet bench, safety glasses, apron, etc.).
- How to handle in case of incidents or accidents

Please note that:

- Most of the safety incidents or accidents are related to working with chemicals.
- Additional Module Training (MT) is required for using chemicals in the labs.

MSDS

Sections in the Material Safety Data Sheet:

- Chemical name and formula.
- Cass number.
- Physical properties.
- Important characteristics.
- Hazards / Symptoms.
- Preventive measures.
- Fire extinguishing / First aid.
- Spillage (how to clean).
- Storage.
- Labelling:
 - Hazard labels.
 - NFPA: Hazard diamond.
 - Risk and Safety (R&S) phrases.
- Remarks / Notes.
- Handling in case of an emergency / Influence on environment.

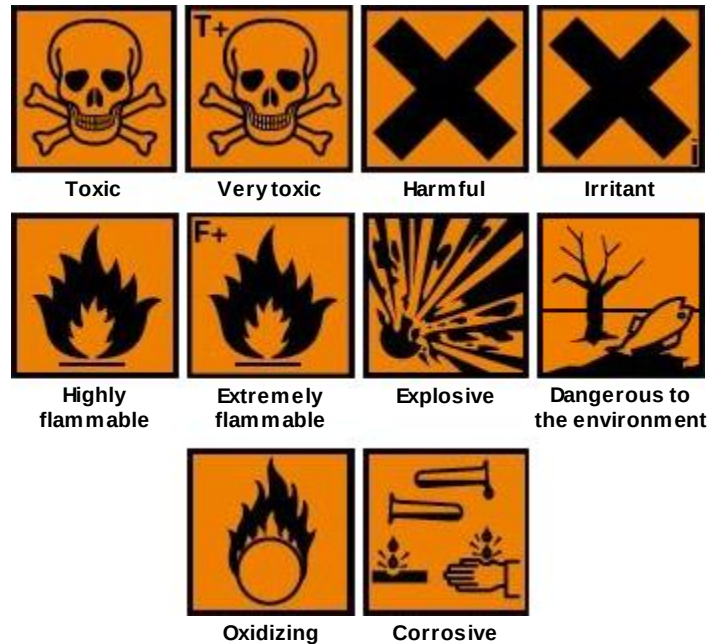
CAS-No: [7697-37-2]		HNO3	
NITRIC ACID (above 70%)			
PHYSICAL PROPERTIES		IMPORTANT CHARACTERISTICS	
Boiling point, °C 83 (100%) Melting point, °C -42 (100%) Relative density (water = 1) 1.5 Relative vapor density (air = 1) 2.2 Relative density at 20 °C of saturated mixture vapor/air (air = 1) 1.06 Vapor pressure, mm Hg at 20 °C 43 (100%) Solubility in water ∞ Relative molecular mass 63.0		COLORLESS TO RED AQUEOUS SOLUTION OF NITRIC ACID IN VARIOUS CONCENTRATIONS ABOVE 70% WITH PUNGENT ODOR Vapor mixes readily with air. Decomposes when heated or exposed to light, giving off toxic gas (→ nitrous vapors). Strong oxidant which reacts violently with combustible substances and reducing agents. Strong acid which reacts violently with bases and is corrosive. Reacts violently with finely dispersed metals, organic solvents and many other substances, with risk of fire and explosion. TLV-TWA 2 ppm 5.2 mg/m³ STEL 4 ppm 10 mg/m³ Absorption route: Can enter the body by inhalation. Harmful atmospheric concentrations can build up very rapidly on evaporation at 20 °C. Immediate effects: Corrosive to the eyes, skin and respiratory tract. Inhalation of vapor/fumes can cause severe breathing difficulties (lung edema). In serious cases risk of death.	
Gross formula: HNO3			
HAZARDS/SYMPTOMS	PREVENTIVE MEASURES	FIRE EXTINGUISHING/FIRST AID	
Fire: non-combustible, but makes other substances more combustible; many chemical reactions can cause fire and explosion. Explosion: many chemical reactions can cause explosion.	avoid contact with combustible substances, solvents and many other substances.	in case of fire in immediate vicinity: use any extinguishing agent. in case of fire: keep tanks/drums cool by spraying with water.	
Inhalation: corrosive; sore throat, cough, shortness of breath, severe breathing difficulties.	STRICT HYGIENE ventilation, local exhaust or respiratory protection.	fresh air, rest, place in half-sitting position, take to hospital.	
Skin: corrosive; redness, pain, burns.	gloves, protective clothing.	remove contaminated clothing, flush skin with water or shower, call a doctor.	
Eyes: corrosive; redness, pain, impaired vision.	face shield, or combined eye and respiratory protection.	flush with water, take to a doctor.	
Ingestion: corrosive; sore throat, abdominal pain.		rinse mouth, immediately take to hospital.	
SPILLAGE	STORAGE	LABELING / NFPA	
collect leakage in sealable containers, soak up with sand or other inert absorbent and remove to safe place, flush away remainder with water, do not use sawdust or other combustible absorbents; (additional individual protection: CHEMICAL SUIT).	keep in cool, dark place, separate from combustible substances, reducing agents and bases; ventilate at floor level.	R: 8-35 S: 23-26-36 Note B  Oxidant  Corrosive 	
NOTES			
In concentrations above 80% is referred to as fuming nitric acid. Physical properties of 100% nitric acid are given. Flush contaminated clothing with water (fire hazard). UN No. of red fuming nitric acid: 2032. Lung edema symptoms usually develop several hours later and are aggravated by physical exertion; rest and hospitalization essential. As first aid, a doctor or authorized person should consider administering a corticosteroid spray. UNDER NO CIRCUMSTANCES add water to acid; when diluting ALWAYS add acid to water. Unbreakable packaging preferred; if breakable, keep in unbreakable container. Packaging: dark glass, aluminum, stainless steel or suitable plastic.			
Transport Emergency Card TEC(R)-9A		HI: 885; UN-number: 2031	

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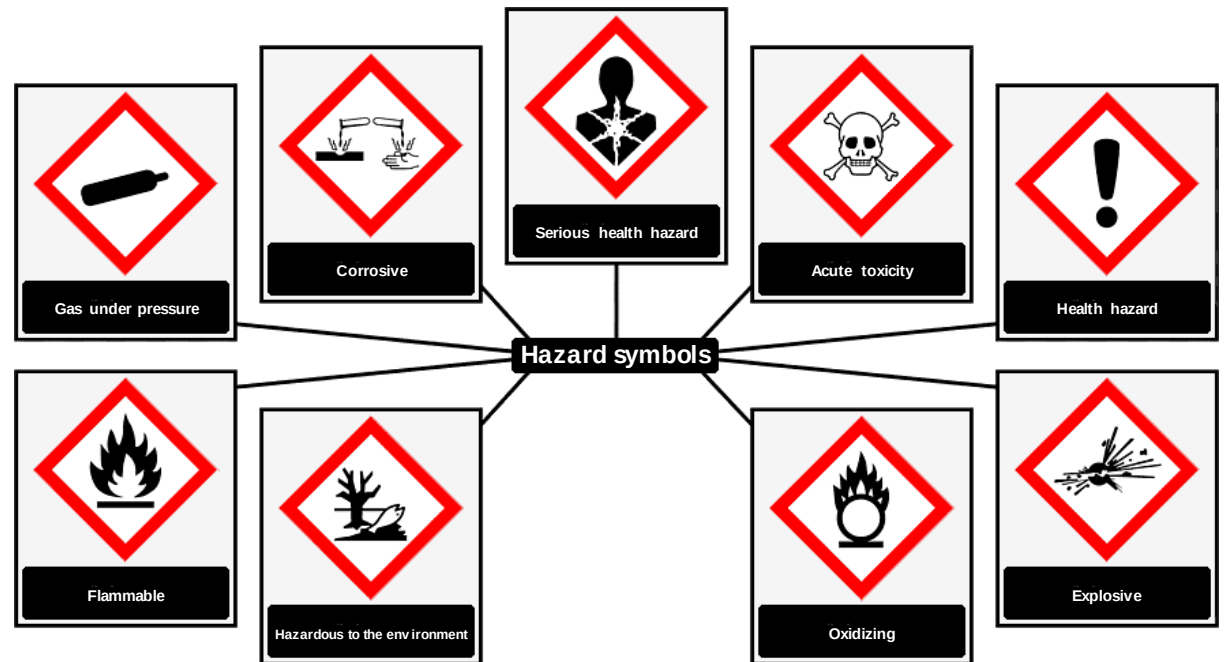
Hazard symbols

The following hazard symbols can be found on bottles containing chemicals, and on some Material Safety Data Sheets.

Old hazard symbols
that were used until 2016.



New GHS (Globally Harmonized System)
symbols, in use since 2016.

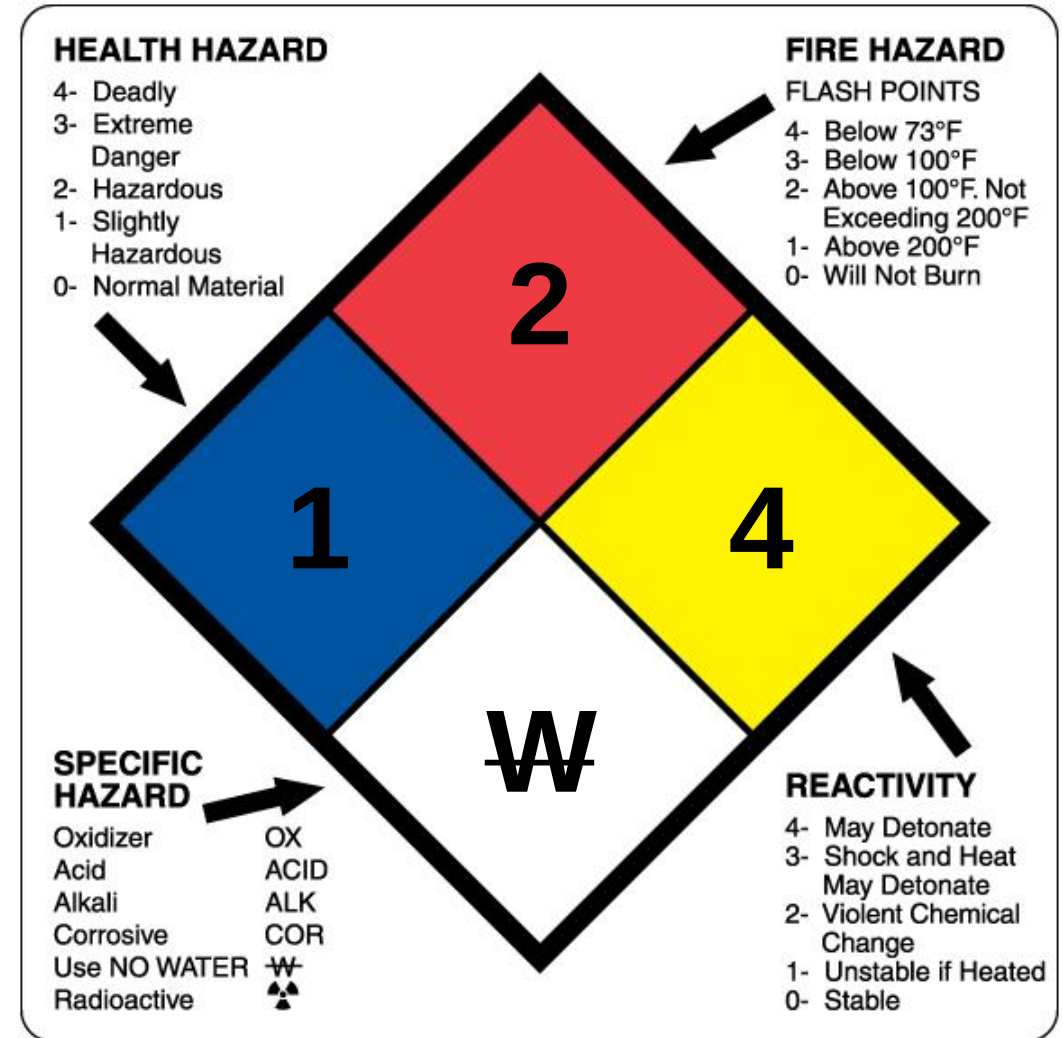


The Hazard Diamond

The **NFPA** (National Fire Protection Agency) diamond shows a quick indication of the hazards and the reactivity of a chemical.

- There are 4 fields:
 - Left (blue) = Health Hazard.
 - Top (red) = Fire Hazard.
 - Right (yellow) = Reactivity.
 - Bottom (white) = Specific Hazard.
- Fire Hazard remarks:
 - 73°F = 22.7°C
 - 100°F = 37.7°C
 - 200°F = 93.3°C

The **flash point** of a flammable liquid is the lowest temperature at which the surface of the liquid emits sufficient vapour to be ignited by a small flame.



Bottle label

A label on a bottle contains the following information:

- Name of the substance (usually in different languages).
- The purity:
 - Technical quality
 - Pro-analyse
 - MOS quality
 - etc..
- The amount.
- The supplier.
- The hazard symbols.



Rules for working with chemicals

- Read the MSDS of used chemicals before you start.
- Use your Personal Protective Equipment (PPE).
- It is mandatory to have a **buddy**.
- Always work in a **fume hood** or **wet bench**.
- Transport chemicals in a proper way to the working area:
 - Use a rubber bottle carrier for 1 bottle or the special trolley for transportation of multiple bottles.



Rubber bucket to transport 1 bottle.



Trolley for transportation of multiple bottles.

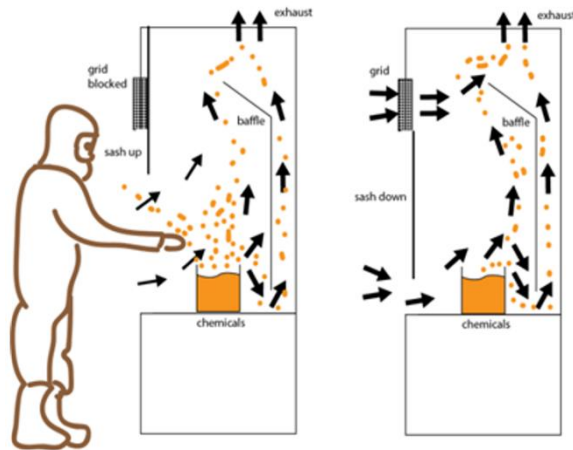
Report every possible hazardous situation to the EKL Safety Officer.

Locations to safely handle/work with chemicals

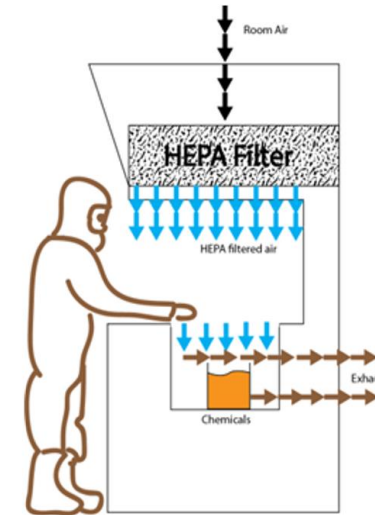
- A safe environment is required to work with chemicals, so that fumes will not be inhaled and accidents can be avoided while using Personal Protective Equipment (PPE).
- Always use the lowest possible area of a wet bench or the sink of a fume hood while pouring chemicals.

EKL provides 2 types of locations:

① a fume hood $\Rightarrow$ does not contain a HEPA filter



② a wet bench $\Rightarrow$ contains a HEPA filter



- EKL has 2 wet bench lines: CR 100 $\Rightarrow$ Mixing and pouring of chemicals is **not** allowed.
 CR 10.000 $\Rightarrow$ Mixing and pouring of chemicals is allowed.

Introducing new chemicals into the EKL labs

- It is strictly forbidden to **ORDER** or **BRING** your own chemicals!
- It is strictly forbidden to create a mix of chemicals that has not been reviewed by your mentor, and is not approved by the EKL Safety Officer.
- If you plan to use a **Carcinogenic, Mutagenic and Reprotoxic (CMR)** chemical, first check if there is an **alternative chemical available that is safer to use.**

Carcinogenic chemicals can cause or promote cancers. Mutagenic chemicals can cause genetic mutations. Reprotoxic chemicals can damage the reproductive process.

Procedure for acquiring new chemicals in EKL:

- Read the MSDS.
- Consult your mentor about the new chemical and your experimental setup.
- Contact the **Wet Bench team** by sending a mail to the **EKL Safety Officer**, with a cc to [Robert Verhoeven](#) and [Anastasiia Kotliarchuk](#). Ask for permission to use the chemical in the EKL cleanroom, and attach the MSDS.
- After approval is granted, contact Robert and Anastasiia and supply the following information:
 - Name of the chemical.
 - Name of the manufacturer.
 - The cost, the amount and the quality (purity).
 - If requested by Robert, then supply the Baan code (financial code).
 - Mention in which lab you plan to use the new chemical.
 - Robert or Anastasiia will order it and provide you with the information in which chemical storage cupboard you can find it. They will also make the MSDS available on the front of the cupboard.

Risk Assessment

What is a Risk Assessment?

The purpose of a risk assessment is to identify hazards in the workplace in order to implement control measures that can eliminate or minimize these risks as much as possible. This will help with providing a safer working environment.

- It's a method to determine the **risks** of activities and workplaces.
- It's goal is to avoid **harm to an individual**.

If a hazard can't be avoided in a safe way, the activity should not be carried out.

Purpose of a Risk Assessment

Why do we need to perform a Risk Assessment?

- To reduce or eliminate hazards.
- To prevent harm to individuals in the lab.
- To define guidelines to create a safer working environment.

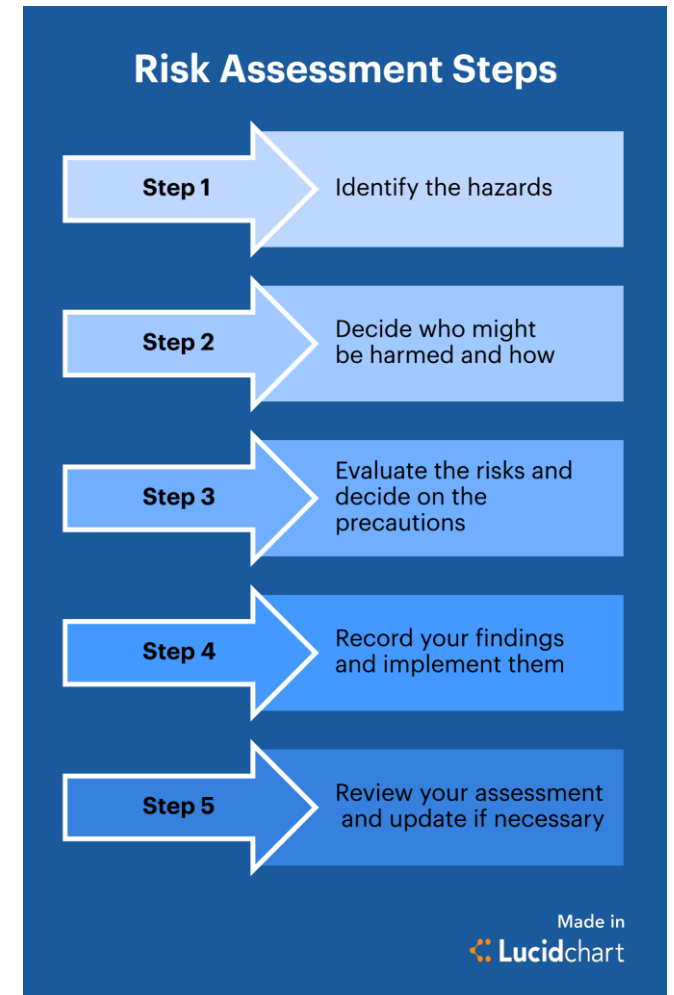
When is a risk assessment required?

- For each use of:
 - New chemicals.
 - A new experimental setup.
 - New equipment.



Steps for writing a Risk Assessment

1. Identify the hazards in the workplace.
The types of hazards you identify depend the workplace or experiment as they are different.
2. Decide who may be harmed and how.
Understanding who could be at risk will help individuals and organizations keep people safe.
3. Evaluate the risks and take action to prevent them.
Evaluate the chance and consequences of risks. Then implement precautions and control measures. E.g. a less risky experimental setup, different chemicals, other equipment, restricted access to areas, providing PPE, giving effective training.
4. Record your findings and implement them.
A written risk assessment provides proof that hazards were evaluated and appropriate action was taken to reduce risk. It is useful for raising awareness about the potential risks of a worksite.
5. Review the assessment and update if necessary.
Work environments are constantly changing. New people come and go, equipment and products are swapped and tested, and new materials are introduced. Make sure risk assessments are up to date.



Result of a Risk Assessment

A risk assessment results in a matrix that contains the chance that a hazard can occur and the consequences that it can have for an individual. When it is created, the safety measures that must be taken have to be defined.

		Likelihood				
		1 Rare	2 Unlikely	3 Possible	4 Likely	5 Almost Certain
Consequences	5 Catastrophic	5 Moderate	10 High	15 Extreme	20 Extreme	25 Extreme
	4 Major	4 Moderate	8 High	12 High	16 Extreme	20 Extreme
	3 Moderate	3 Low	6 Moderate	9 High	12 High	15 Extreme
	2 Minor	2 Low	2 Moderate	6 Moderate	8 High	10 High
	1 Negligible	1 Low	2 Low	3 Low	4 Moderate	5 Moderate

Risk Assessment procedure

- A training course given by the safety officer can be requested in NIS.
- EKL has a **template form** on SharePoint with guidelines for writing a risk assessment.
- If needed ask for help from your mentor or a tool owner (for equipment related questions).
- Fill out the form and send it to the Safety Officer.

TU Delft		Safety Report Sheet		Version name: EEMCS-052023 Version number: V.0-2022 Page 1 of 8								
Experiment Title: _____												
Report number & Date: _____												
Name of the experiment unit: _____												
User/Author & e-mail: _____												
Section/Research Group: _____												
Location: _____												
Supervisor/Approval: _____												
Area Supervisor: _____												
Review by: _____												
Researcher Phone Number in case of Emergency: _____												
1.1 Experiment Objective: _____												
1.2 Hazardous Materials Used (chemicals, gases, biological agents) & risk groups:												
#	Material Name & formula	CAS#	Chemical state (solid, liquid, gas)	Quantity	Unit	Chemwatch Hazard Ratings*						
1						<table border="1"> <tr> <td>Flammability</td> <td>Reactivity</td> </tr> <tr> <td>Toxicity</td> <td>Chronic</td> </tr> <tr> <td>Body contact</td> <td></td> </tr> </table>	Flammability	Reactivity	Toxicity	Chronic	Body contact	
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Nano-particles

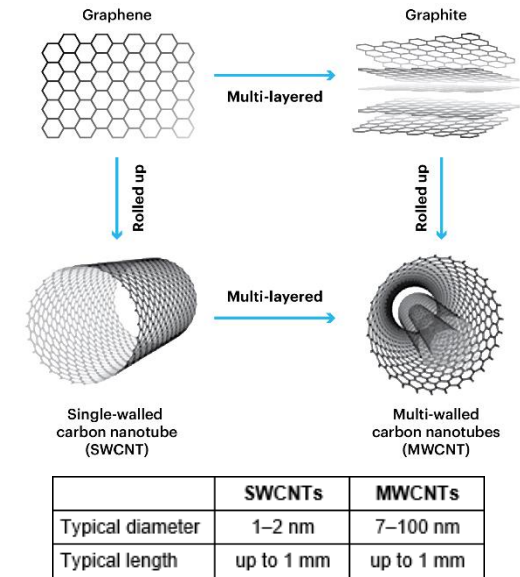
Nano-particles are particles that have at least 1 dimension between 1-100 nm.

Not a lot of knowledge is available about the unique properties of engineered nano-materials and what the influence of them is on peoples health. Always assume the **worst**! Due to the large surface area they are highly reactive.

Possible health hazards:

- Absorption deep into the lungs.
- Migration through the skin.
- A delayed reaction.

You need **special permission** to work with nano-particles from the [EKL Process Coordinator](#).



Check the EKL SharePoint site for the procedure:

https://sharepoint.tudelft.nl/sites/EKL2/Safety/_layouts/15/WopiFrame.aspx?sourcedoc=/sites/EKL2/Safety/Shared%20Documents/Nanoparticles.pdf&action=default

Reporting of incidents or accidents, and improvements & suggestions

Create a report using the template on SharePoint:

<https://sharepoint.tudelft.nl/sites/EKL2/Safety/SitePages/Home.aspx>

Click the button to report an incident or accident!



- All incidents or accidents **must** be reported. They will be analyzed and an action list will be defined.
- You can follow the status of your reported incident on:
<https://sharepoint.tudelft.nl/sites/EKL2/Safety/IncidentsFiles/Forms/MyIncidents.aspx>

